Venkata Lakshmi Parimala Pasupuleti

Applied AI Engineer · Customer-Facing AI · LLMs in Production · End-to-End Delivery · FastAPI · Python · AWS · React

venkatalakshmiparimala@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |
AgentProbe (live eval toolkit) | Fairfax, Virginia · Open to Onsite NYC

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production experience. I build the factory, not just what runs on it: at Intuit I built the LLM evaluation framework, guardrail architecture, and RAG grounding pipeline that financial AI features run on top of -- not the features themselves. Built **Leasel**: an 8-agent LangGraph platform where every answer is **source-backed** (100% citation correctness, 0% unsupported-claim rate), **permissioned** by role and context (tenant safety boundary adversarially tested -- never resolves a regulated decision without human sign-off), and **reliable enough to inform a compliance decision** (93.6% precision, 87.9% F1 across 1,240 test cases). Built **AgentProbe**: open-source LLM evaluation toolkit running across 5 providers -- extreme ownership, built solo from scratch. 7+ years at Entrata (100K+ MAU SaaS) owning complex, ambiguous problems end-to-end: from an underdefined regulatory requirement to a production system that has run for 7 years without a compliance violation. Production-ready code across Python, FastAPI, Java, React, PostgreSQL, AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Customer-Facing AI Solutions | LLMs in Production | End-to-End Delivery | AI Agents & Agentic Systems | RAG Workflows & Retrieval | Evals & LLM-as-Judge | Workflow Orchestration | High Autonomy Prototyping | Python & FastAPI | React & Full-Stack | AWS & Docker | Technical Roadmap Ownership

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — LLMs in Production · Customer-Facing AI · FastAPI · Python · AWS

- Own **customer-facing AI features end-to-end**: discover the use case, build **LLM applications** (Anthropic Claude) with Python/FastAPI, ground responses in customer financial data via **RAG workflows**, curate the technical roadmap from prototype to production -- not handed a spec, responsible for the whole loop including iteration post-launch.
- Build **LLM evaluation harness** with LLM-as-judge scoring and golden datasets; CI-integrated regression detection caught 3 prompt regressions before release, zero hallucination incidents post-deployment; add guardrails (input validation, sensitive-data redaction, fallback paths) at the code layer for safe production AI.
- Build ML pipelines (scikit-learn, XGBoost) for financial predictions; develop React/TypeScript front ends; containerize services with Docker, ship through GitHub Actions to AWS with OpenTelemetry monitoring and logging.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · React · AWS · PostgreSQL

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints; **curated technical roadmaps** cross-functionally with Legal, Compliance, Operations, and Product -- part PM, part engineer on every delivery; reviewed 300+ PRs; mentored junior engineers.
- Built **Python/Bash automation tools** eliminating 80% of manual operational work; built internal data investigation tools cutting triage time 40%; optimized 200+ SQL queries (30% gain) via execution plan analysis and indexing.
- Owned HUD income-limit ETL pipeline for 7 years: few thousand federal rows → 180K domain records with validation, rollback checks, and compliance enforcement -- zero violations; RabbitMQ **workflow orchestration** with idempotency guarantees for bulk tenant record processing.
- AWS S3 PHI migration (Libsodium encryption, 100% PHI compliance, zero security incidents); Docker containerization cutting microservices execution time 50%.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · PyTorch · HuggingFace · Statistical Modeling

- Built predictive models achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection; built fine-tuned classification model (HuggingFace + PyTorch, presented to faculty panel). Mentored 600+ students across Python and database design.

PORTFOLIO PROJECTS

Leasel — Customer-Facing AI Platform with End-to-End Delivery

Customer-Facing AI | LLMs in Production | Agentic AI

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG, SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Owned **end-to-end delivery** across 3 customer-facing surfaces (Document Intelligence & Review, Support Ticket Investigation, Tenant Self-Service) sharing one SQLite knowledge repo -- curated **customer-specific technical roadmap** for each surface independently; part PM, part engineer throughout.
- 8-agent LangGraph StateGraph with real conditional edges and **confidence-gated HITL escalation**: triage → retrieval → config/history/structured-data agents → reasoning → validation → human; correctly traces false-ineligibility ticket to misconfigured setting tagged to specific release.
- **RAG workflow** (hand-rolled TF-IDF + cosine reranking): 100% citation correctness, 0% unsupported-claim rate; AgentProbe **eval harness**: 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe — Open-Source LLM Evaluation Toolkit

Evals | LLM-as-Judge | Multi-Provider

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built end-to-end from scratch with **high autonomy**: 15-endpoint FastAPI backend, LLM Router for dynamic task routing across 5 providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama), 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix), ChromaDB RAG evaluation -- 100% local, zero vendor lock-in.
- **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval quality, completeness, compliance); CI-integrated regression detection; used to evaluate Leasel across 1,240+ test cases.

AI on FHIR — Clinical NLP with Customer-Facing AI & HIPAA Guardrails

Customer-Facing AI | Rapid Prototype

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish clinical questions into FHIR API calls using LLMs, BERT, and embeddings; HIPAA guardrails (RBAC, PHI output filtering); ~60% reduction in manual lookup time; containerized with Docker for rapid deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI & LLMs: LLMs in Production, LLM Applications, AI Agents, Agentic AI, Customer-Facing AI, RAG Workflows, Retrieval-Augmented Generation, Evals, LLM-as-Judge, Guardrails, HITL, Responsible AI, Prompt Engineering, Structured Outputs, LangGraph, LangChain, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Hallucination Detection

Retrieval & NLP: RAG Pipelines, Hybrid Search, Similarity Search, ChromaDB, Pinecone, Weaviate, Elasticsearch, TF-IDF, Embeddings, Semantic Search, BERT, HuggingFace, spaCy, Document Intelligence, OCR

Python & Full-Stack: Python (9+ yrs), FastAPI, Flask, Django, React, TypeScript, Node.js, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark, Java, Spring Boot, PHP, SQL, PL/SQL, Bash

Workflow & Data: Workflow Orchestration (RabbitMQ, Celery, Temporal-adjacent), dbt-adjacent ETL, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, Data Modeling, Query Optimization

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana